



University
of Exeter

Text Mining & NLP

COMM040

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Intended Learning Outcomes

- Understand the strengths, weaknesses and biases of different NLP methods
- Clean, parse, transform and process large collections of text documents
- Use modern NLP techniques to derive insight from unstructured data
- Use modern NLP techniques to derive insight from unstructured data

Outline

Recap & Motivation

Keywords

Part of Speech (PoS)

Syntax

Outline



Recap & Motivation

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Syntax

Last week: ILO

- Understand the strengths, weaknesses and biases of different NLP methods
 - Analyse how ambiguity and representation choices influence NLP methods, limitations, and biases.
- Clean, parse, transform and process large collections of text documents
 - Apply tokenisation and encoding-aware text cleaning to prepare textual datasets.
- Know about recent advances and applications of NLP methods, their impact and real-world applications.
 - Explain core NLP applications and their relevance to contemporary real-world systems.
- Use modern NLP techniques to derive insight from unstructured data
 - Generate lexical statistics and evaluate tokenisation strategies.

Tokens are NOT enough

- How can we encode a text?
- [menti.com](https://www.menti.com)

Tokens are NOT enough

- How can we encode a text?
- [menti.com](https://www.menti.com)
- **But frequency is not enough**
 - ambiguity
 - sparsity
 - lack of structure

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What Are Keywords?

- **Keywords** are words or short phrases that represent the **main topics** of a text.
- They provide a **compact semantic summary** of a document.
- Widely used for:
 - Information Retrieval
 - Document Classification
 - Summarisation
 - Metadata generation

Keywords vs Other Text Units

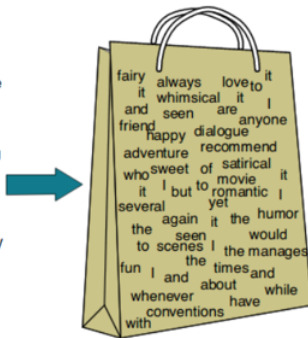
- **Tokens:** all word units in the text
- **Stopwords:** frequent but low-information words
- **Keywords:** salient and informative tokens/words
- **Important:**
 - Not all frequent words are keywords
 - Not all keywords are frequent
 - Keywords from a text vs keywords from a domain

How Are Keywords Extracted?

- **Statistical approaches**
 - Frequency-based methods
 - TF-IDF
 - Association measures (e.g., Log-likelihood)
- **Graph-based approaches**
 - Word co-occurrence graphs
 - Centrality-based ranking
- **Model-based approaches**
 - Embeddings
 - Language models

Bag-of-Words (BoW)

I love this movie! It's sweet,
but with satirical humor. The
dialogue is great and the
adventure scenes are fun...
It manages to be whimsical
and romantic while laughing
at the conventions of the
fairy tale genre. I would
recommend it to just about
anyone. I've seen it several
times, and I'm always happy
to see it again whenever I
have a friend who hasn't
seen it yet!



it	6
I	5
the	4
to	3
and	3
seen	2
yet	1
would	1
whimsical	1
times	1
sweet	1
satirical	1
adventure	1
genre	1
fairy	1
humor	1
have	1
great	1
...	...

Bag-of-Words (BoW)

- Vectorisation with BoW normally uses token frequency (aka term frequency)
- For multiple documents, need to use the full vocabulary (union of unique tokens)

Document 1: "the cat sat on the mat"

Document 2: "the cat sat"

Unique unigrams: the, cat, sat, on, mat

Bag-of-words 1: {"the": 2, "cat": 1, "sat":1, "on":1, "mat":1}

Bag-of-words 2: {"the":1, "cat":1, "sat":1}

Term frequency vector 1: [2,1,1,1,1]

Term frequency vector 2: [1,1,1,0,0]

Normalization

1. Putting words/tokens in a standard format:
 - U.S.A. or USA
 - uhhuh or uh-huh
 - am, is, be, are
2. Case folding: *reduce all letters to lower case*
3. Lemmatization: represent words as their lemma (shared root):
 - am, are, is → be;
 - ant, wants → want;
 - *Spanish* quiero ('I want'), quieres ('you want') → querer
'want'

Weaknesses of Term Frequency (TF)



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- Frequency $\neq$ importance.
- High-frequency words dominate counts.
- Frequent words (the, and, of) are not content-bearing.
- Document-specific frequency ignores global distribution.
- Common terms overwhelm informative ones.

Document Weighting

- A rare term appearing once may be more informative than a frequent one.
- Need a principled way to measure informativeness across a corpus.

Document Weighting

- Inverse Document Frequency (IDF)
- $\text{IDF}(w) = \log\left(\frac{N}{\text{df}(w)}\right)$
 - N = number of documents
 - $\text{df}(w)$ = number of documents containing w
 - Rare terms receive higher weights.

Document Weighting

- TF-IDF
- $\text{TF-IDF}(w, d) = \text{TF}(w, d) \times \text{IDF}(w)$
 - Weighted lexical representation
 - Sparse but interpretable
 - Backbone of classical information retrieval

Document Weighting

- Questions about TF-IDF
- [menti.com](https://www.menti.com)

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Syntax

Syntax is Sequential

- Word order matters
 - “cats chase mice” $\neq$ “mice chase cats”
- PoS tags form structured sequences
- Syntax expresses preferences between neighbouring tags

Ambiguity Requires Context

- Lexically ambiguous words
 - *book*
 - *flies*
 - *can*
- Surrounding words enable disambiguation
 - *Time flies like an arrow.*
 - *Can you can a can?*

Part-of-speech tagging

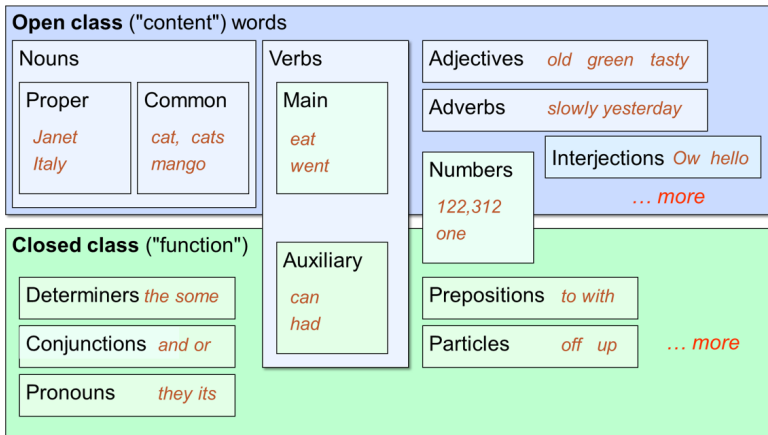
- Words with similar grammatical

Noun	Pronoun
Refers to a person, concept, place, or thing	Used in place of a noun
Adjective	Adverb
Modifies a noun or pronoun	Can modify a verb, adjective, adverb, or whole sentence
Verb	Preposition
Describes an action, occurrence, or state of being	Used to show the relationship between the different parts of a sentence
Conjunction	Interjection
Connects different parts of a sentence	Used in isolation to express a feeling, give a command, or greet someone

Color-coded example sentence:

Well, we like to visit old friends now and then.

Part-of-speech tagging



Part-of-Speech Tagging

- Example:

There/**PRO** were/**VERB** 70/**NUM** children/**NOUN**
there/**ADV** ./PUNC

Preliminary/**ADJ** findings/**NOUN** were/**AUX**
reported/**VERB** in/**ADP** today/**NOUN** 's/**PART**

New/**PROPN** England/**PROPN** Journal/**PROPN** of/**ADP**
Medicine/**PROPN**

Part-of-Speech Taggers

- Historical Evolution:
 - Rule-based systems (1980s)
 - HMM taggers (1990s)
 - CRFs and LSTMs (2000s)
 - Transformers (2017+)

Rule-based systems PoS tagger

- A tag set and a large lexicon + handcrafted rules.
- Rules look at:
 - word suffixes (e.g., -ing, -ed, -ly)
 - capitalisation
 - orthographic patterns (digits, hyphens)
 - neighbouring tags

Rule-based systems PoS tagger

- **Lexical rules**

- If token has only one tag → assign it.
- If ending in `-ness` → **NOUN**.
- If ending in `-able` → **ADJ**.

- **Contextual (Constraint) rules**

- If word = `to` and next word is VB base form → tag as **TO**.
- If previous word is a determiner → prefer **NOUN** over **VERB**.

- **Correction rules**

- Re-evaluate tags based on later context.

Rule-based systems PoS tagger

- Propose PoS rules
- [menti.com](https://www.menti.com)

Rule-based systems PoS tagger

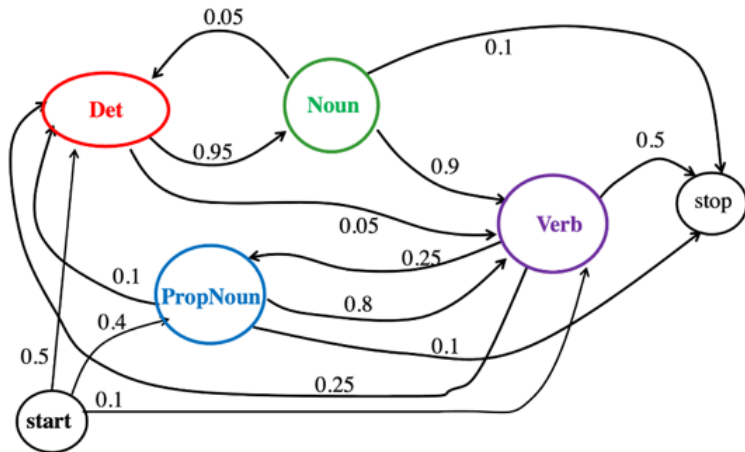
Strengths

- Transparent and interpretable decisions.
- High precision when domain is constrained.
- Useful in early NLP when annotated corpora were scarce.

Weaknesses

- Expensive to build and maintain.
- Weak outside the domain they were written for.
- Hard to capture ambiguity statistically.

Part-of-Speech Tagging (HMM)



Part-of-Speech Tagging

- HMM (Hidden Markov Model)
 - Hidden states
 - ◊ Part-of-speech tags
 - Observations
 - ◊ Words
 - Probabilities
 - ◊ Transition probabilities encode syntax
 - ◊ Emission probabilities encode lexical preferences

Part-of-Speech Tagging (HMM)

- Learned automatically from corpus data
 - High-probability transitions
 - ◊ $P(\text{NOUN} \mid \text{DET})$
 - Low-probability transitions
 - ◊ $P(\text{VERB} \mid \text{DET})$
- Enables probabilistic disambiguation
 - $P(\text{bank} \mid \text{NOUN})$
 - $P(\text{bank} \mid \text{VERB})$

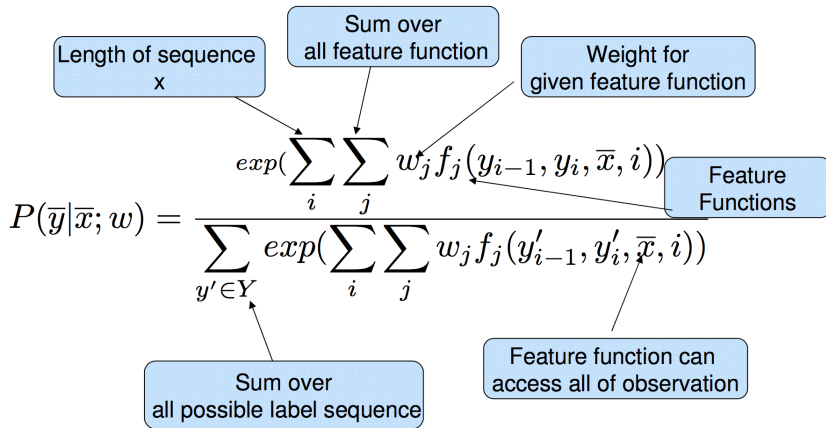
Part-of-Speech

- Limitations of HMMs
 - Strong independence assumptions
 - Cannot incorporate rich contextual features
 - Poor modelling of long-distance dependencies

What Is a CRF?

- Conditional Random Fields (CRFs)
- A **discriminative** sequence model.
- Directly models: $P(\text{tags} \mid \text{words})$
- Allows arbitrary, overlapping, and dependent features:
 - word identity
 - suffixes/prefixes
 - capitalization, digit patterns
 - linguistic cues (shape, POS lexicons)
- No need to model $P(\text{words})$ (unlike HMMs).

What Is a CRF?



The diagram illustrates the CRF probability formula with several annotations in blue boxes:

- Length of sequence x** : Points to the $\bar{x}$ term in the feature function arguments.
- Sum over all feature function**: Points to the inner summation over j .
- Weight for given feature function**: Points to the w_j term.
- Feature Functions**: Points to the f_j term.
- Feature function can access all of observation**: Points to the $\bar{x}$ term in the denominator's feature function arguments.
- Sum over all possible label sequence**: Points to the outer summation over $y' \in Y$.

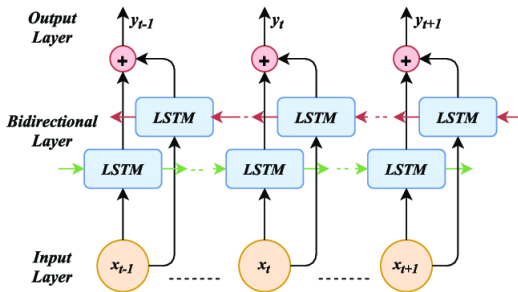
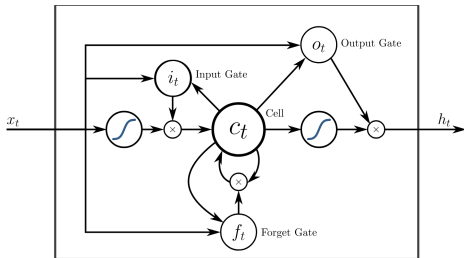
$$P(\bar{y}|\bar{x}; w) = \frac{\exp\left(\sum_i \sum_j w_j f_j(y_{i-1}, y_i, \bar{x}, i)\right)}{\sum_{y' \in Y} \exp\left(\sum_i \sum_j w_j f_j(y'_{i-1}, y'_i, \bar{x}, i)\right)}$$

Why CRFs Outperform HMMs

- CRFs use feature functions such as:
 - $f_1(\text{word}_t = \text{"ing"}, \text{tag}_t = \text{VBG})$
 - $f_2(\text{isCapitalized}, \text{tag}_t = \text{NNP})$
 - $f_3(\text{tag}_{t-1} = \text{DET}, \text{tag}_t = \text{NOUN})$
- Advantages:
 - No independence assumptions on features.
 - Allows complex and long-range correlations.

LSTM

- Long Short-Term Memory (LSTM)



LSTMs for PoS Tagging

- PoS decisions depend strongly on context:
 - *book* (NOUN) vs. *book* (VERB)
- LSTMs learn contextual patterns directly from data.
- The model builds an evolving representation of the sentence.
- No need for handcrafted features (suffixes, tag windows, etc.).
- Ideal for capturing short-range syntactic behaviour.

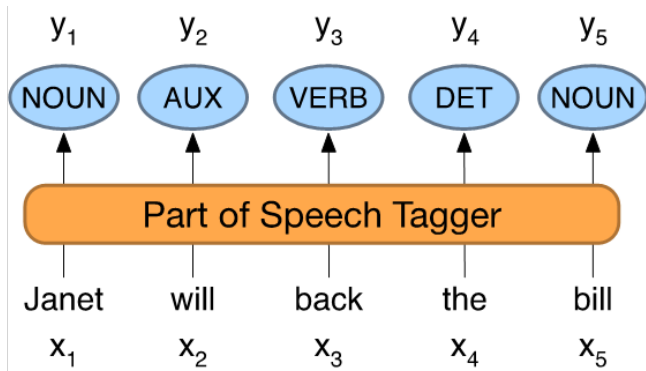
How an LSTM Works

- Reads the sentence left to right.
- Maintains a **memory state** summarising relevant context.
- Learns what to keep and what to forget.
- Combines memory + current word embedding to predict the PoS tag.

LSTMs for PoS Tagging

- Models short-range syntax robustly (DET $\rightarrow$ ADJ $\rightarrow$ NOUN).
- Disambiguates using local context.
- Learns useful morphological cues via embeddings.
- Handles variable-length contexts without manual feature design.
- Not ideal for very long dependencies. PoS rarely requires them.

Part-of-Speech Using Transformers



Part-of-Speech

- Current PoS approach:
 - Conditional Random Fields (CRFs)
 - Transformer-based models

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Examples of Tag Sequences

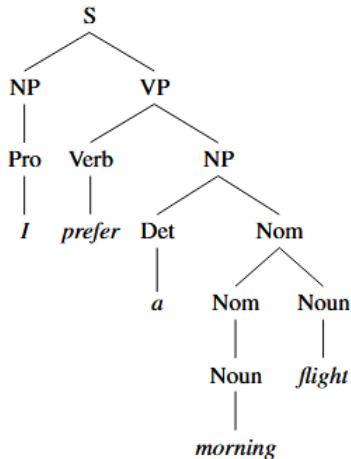
- Common transition
 - $DT \rightarrow NN$
- Rare transition
 - $DT \rightarrow VB$
- What are common transitions in English grammar?
 - [menti.com](https://www.menti.com)

What Is Syntactic Parsing?

- Parsing assigns a **structure** to a sentence.
- Words group into constituents (NP, VP, PP, ...).
- A parser builds a tree showing how phrases combine.
- Even modern neural models implicitly learn these structures.

Constituency Parsing

Grammar Rules	Examples
$S \rightarrow NP VP$	I + want a morning flight
$NP \rightarrow$ <i>Pronoun</i>	I
<i>Proper-Noun</i>	Los Angeles
<i>Det Nominal</i>	a + flight
<i>Nominal</i> $\rightarrow$ <i>Nominal Noun</i>	morning + flight
<i>Noun</i>	flights
$VP \rightarrow$ <i>Verb</i>	do
<i>Verb NP</i>	want + a flight
<i>Verb NP PP</i>	leave + Boston + in the morning
<i>Verb PP</i>	leaving + on Thursday
$PP \rightarrow$ <i>Preposition NP</i>	from + Los Angeles



Shift-Reduce Parsing

- 1: **Input:** sentence $w_1 \dots w_n$
- 2: $\text{stack} \leftarrow []$
- 3: $\text{buffer} \leftarrow [w_1 \dots w_n]$
- 4: **while** not finished **do**
- 5: **if** top of stack matches a grammar rule **then**
- 6: pop Right-Hand Side symbols from stack
- 7: push resulting constituent
- 8: **else**
- 9: pop next token from buffer
- 10: push token onto stack
- 11: **end if**
- 12: **end while**
- 13: **Return** the remaining stack element

Shift-Reduce Parsing

Shift-Reduce Parsing (Step 1)

Stack: []	Buffer: [The, cats, chase, mice]	Action: SHIFT
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(no partial tree yet)

Shift-Reduce Parsing

Shift-Reduce Parsing (Step 2)

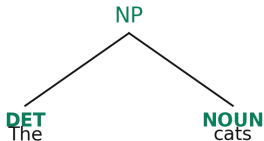
Stack: [The/DET]	Buffer: [cats, chase, mice]	Action: SHIFT
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DET
The

Shift-Reduce Parsing

Shift-Reduce Parsing (Step 3)

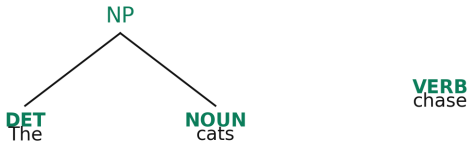
Stack: [NP]	Buffer: [chase, mice]	Action: REDUCE → NP
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Shift-Reduce Parsing

Shift-Reduce Parsing (Step 4)

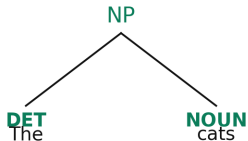
Stack: [NP, chase/VERB]	Buffer: [mice]	Action: SHIFT
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Shift-Reduce Parsing

Shift-Reduce Parsing (Step 5)

Stack: [NP, chase/VERB, mice/NOUN]	Buffer: []	Action: SHIFT
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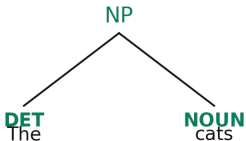
VERB
chase

NOUN
mice

Shift-Reduce Parsing

Shift-Reduce Parsing (Step 6)

Stack: [NP, chase/VERB, NP]	Buffer: []	Action: REDUCE → NP
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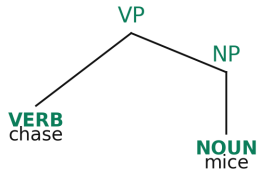
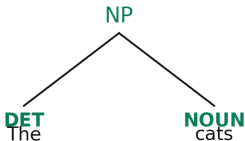
VERB
chase



Shift-Reduce Parsing

Shift-Reduce Parsing (Step 7)

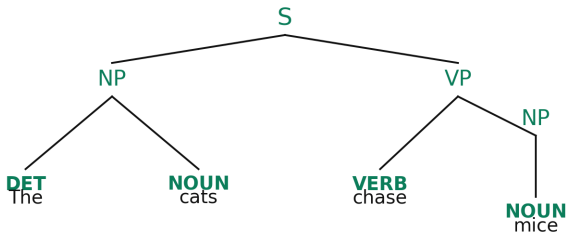
Stack: [NP, VP]	Buffer: []	Action: REDUCE → VP
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Shift-Reduce Parsing

Shift-Reduce Parsing (Step 8)

Stack: [S]	Buffer: []	Action: REDUCE → S (DONE)
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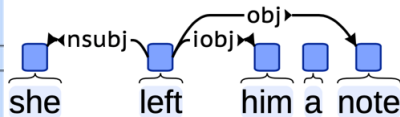


Question

- You are parsing the sentence
 - “The cats chase mice”
- Current configuration:
 Stack [DET, NOUN]
 Buffer [chase, mice]
- What is the next correct action? [menti.com](https://www.menti.com)

Dependency parsing

Clausal Argument Relations	Description
NSUBJ	Nominal subject
OBJ	Direct object
IOBJ	Indirect object
CCOMP	Clausal complement
Nominal Modifier Relations	Description
NMOD	Nominal modifier
AMOD	Adjectival modifier
APPOS	Appositional modifier
DET	Determiner
CASE	Prepositions, postpositions and other case markers
Other Notable Relations	Description
CONJ	Conjunct
CC	Coordinating conjunction



Dependency parsing

- Universal Dependencies project

<https://universaldependencies.org/>

Conclusion

- By the end of this session, students will be able to:
 - Compare frequency-, statistical-, and sequence-based methods in terms of expressiveness, limitations, and bias.
 - Apply weighting, keyword extraction, and syntactic annotation to structured text representations.
 - Explain methods of keyword extraction, PoS tagging, and parsing
 - Demonstrate methods of language understanding.

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